

The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane

$$\operatorname{Re}(s) > 7/8 - 5 \times 10^{-6}$$

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Abstract

We examine the numerical endpoint in the compensated-probe proof of the seven-eighths zero-free half-plane in [1]. The reflected-row estimate has room to increase the total prime length, while the high-side endpoint certificate has a fixed positive margin. We extend the source moment estimates to the perturbed geometry and obtain the zero-free boundary $174999/200000 = 0.874995$. The accompanying Lean proof derives the moment input and the continuation conclusion from the upstream formalization.

1 Statement and geometry

Let $F = \mathbb{Q}(\sqrt{-3})$. As in [1, Equation (1.1b)], let β_* be the supremum of $1/2$ and the real parts of zeros of primitive finite-order Hecke L -functions over F in the critical strip. We use the seven-eighths theorem of [1], so $\beta_* \leq 7/8$. We also use the arithmetic identities, detector estimates, reflected-energy lemma, additive Gram bound, and continuation criterion proved there. Section 2 checks the changed low comparison. Section 3 supplies the perturbed source moments with zero moment excess, and Section 4 checks the high comparison and analytic domains.

Theorem 1. *Every finite-order Hecke L -function over F and every Dirichlet L -function, including the Riemann zeta function, has no zero in*

$$\operatorname{Re} s > B := \frac{174999}{200000} = \frac{7}{8} - \frac{1}{200000}.$$

The usual pole at $s = 1$ for a principal character is allowed.

Put

$$\begin{aligned} t &= \frac{3}{100000}, & \ell &= \frac{1}{6} + t, & b &= \frac{1}{8}, \\ l_x &= \frac{17}{48} - \frac{t}{2}, & l_y &= \frac{23}{48} - \frac{t}{2}, & h &= \frac{13}{16} + \frac{3t}{2}, \\ M &= l_x + l_y = \frac{5}{6} - t. \end{aligned} \tag{1}$$

These lengths satisfy $l_y - l_x = b$, $M + \ell = 1$, and $h = 1 - l_x + \ell$. The general Mellin-exponent identity in [1, Part II, compensated low estimate] therefore remains

$$C(s) = \frac{l_x}{2} + s - 1 + \frac{h}{6} = s - \frac{11}{16}. \tag{2}$$

Choose finitely many distinct positive slot lengths ℓ_j with $\sum_j \ell_j = \ell$. Their maximum can be made smaller than any prescribed positive mesh. The prime annuli at scales Z^{ℓ_j} are then pairwise disjoint for all sufficiently large Z . The finite physical probe and its exact Poisson identity in [1] make sense with the lengths in (1). All its arithmetic factors are unchanged.

2 The low comparison

Lemma 2. *The normalized compensated probe of [1], formed with (1), satisfies for every $\varepsilon > 0$*

$$|J_\eta(Z)| \ll_{\eta,\varepsilon} Z^{3/16-t/4+\varepsilon}.$$

Proof. Fix a rescaled subset of slots of total logarithmic length $0 \leq d \leq \ell$. In the notation of the compensated low estimate of [1], its row and surviving-slot lengths are

$$M' = M - 2d, \quad \ell' = \ell - d.$$

The proof of the reflected-row norm in [1, Part II, compensated low estimate] first bounds one branch by M' and the other by $M' + (1 + 3\ell' - 2M')/4$, up to arbitrary small losses. Its general reflected-energy lemma permits these fixed bounded lengths. Substitution gives the squared row norm

$$\sum_{0 < N \leq m \leq Z^{M'}} |B_{m,\sigma}^J(Z)|^2 \ll Z^{M' + ((d-1/6+5t)/4)_+ + \varepsilon}. \quad (3)$$

The additive Gram estimate is available throughout this range. Its scale conditions follow from

$$\begin{aligned} l_x - \ell &= \frac{3}{16} - \frac{3t}{2} > 0, & l_y - \ell &= \frac{5}{16} - \frac{3t}{2} > 0, \\ M - 2\ell &= \frac{1}{2} - 3t > 0, & l_y - \ell - \frac{11b}{6} &= \frac{1}{12} - \frac{3t}{2} > 0. \end{aligned}$$

Thus its P_a^2/Y' term is absorbed by $P_a^{1/6}$, exactly as in [1, Part II, additive Gram bound]. Cauchy–Schwarz, the count of rescaled tuples, and their coefficients add to the basic low exponent

$$-d + \left(\frac{d - 1/6 + 5t}{8} \right)_+.$$

This is nonpositive for $0 \leq d \leq \ell$. Indeed, its positive-part argument is negative until $d > 1/6 - 5t$, and thereafter $d - 1/6 + 5t < 8d$. The empty subset has exponent zero. The direct bound is consequently $Z^{l_x/2+b/12+\varepsilon}$.

The prime normalizer in [1, Part II, principal normalizer] is nonzero and has inverse $Z^{o(1)}$ for every fixed positive slot length. Finally,

$$\frac{l_x}{2} + \frac{b}{12} = \frac{3}{16} - \frac{t}{4}.$$

This proves the normalized estimate. □

The signal at the proposed boundary is

$$C(B) = \frac{3}{16} - \frac{1}{200000}, \quad C(B) - \left(\frac{3}{16} - \frac{t}{4} \right) = \frac{1}{400000} > 0. \quad (4)$$

3 The perturbed source moments

The fourth-moment lemma with short prime factors in [1] permits $\kappa = 3/4$. Its zero-free hypothesis is available because $2\beta_* - 1 \leq 3/4$. The original detector application uses $\kappa = 3/4 + 2\Delta$ with $\Delta = \beta_* - 7/8 > 0$. Here the moment estimates use $\Delta = 0$. The gap $\beta_* - B$, when positive, is used separately to choose the detector losses.

We record the estimates needed for this application. Use the normalized witness polynomials and physical slot factors of [1]. In a source fiber $\mathcal{B}$ at scale $U = Z^d$, write r and m for the inverse and plain witness lengths, $I_{u,n}$ for the inverse witness with n logarithmic derivatives, and $P_{u,j}$ for the plain witness with j logarithmic derivatives. These include the real shifts in $[0, 1]$ and the height shifts in $[-H, H]$. The physical slot factors use $\operatorname{Re} z = 17/50$ and $|\operatorname{Im} z| \leq H$. For a selected set A of slots, let $Q_A(u)$ be their physical product and put

$$\lambda_j = \ell_j/d, \quad L_A = \sum_{j \in A} \lambda_j.$$

Lemma 3. *Suppose $B < \beta_* \leq 7/8$. For every admissible choice of the perturbed detector losses, let $\rho > 0$ be its moment loss. There is a positive mesh μ such that slot lengths with $\ell_j \leq \mu/200$ satisfy the following estimates. After fixing the slots and source, there is an exponent $J \geq 0$ independent of the target character. For each target there are $C_\eta > 0$ and a threshold such that the bounds hold for every source fiber with $1/200 \leq d \leq 7/8$, $51/100 < a \leq 7/8$, and retained rows $Z^{1/100} \leq Nu \leq Z^{d-\rho}$ above that threshold. With $C_H = C_\eta(1+H)^J$, uniformly for $H \geq 0$,*

$$\sum_{u \in \mathcal{B}} |I_{u,n} Q_A(u)|^2 \leq C_H U^{1+\rho}, \quad n \leq 2, \quad r + 2L_A < 1, \quad 2r + 8L_A < 3, \quad (5)$$

$$\sum_{u \in \mathcal{B}} |P_{u,j} P_{u,k} Q_A(u)|^2 \leq C_H U^{1+\rho}, \quad j + k \leq 2, \quad 2m + \frac{9}{2}L_A \leq 1, \quad (6)$$

$$\sum_{u \in \mathcal{B}} |P_{u,j} P_{u,k}|^2 \leq C_H U^{\max\{1, 2m\} + \rho}, \quad j + k \leq 2. \quad (7)$$

The unmarked inverse raw-moment bounds also hold with the count parameters chosen from the fixed source. Together these are the source moment bounds with zero moment excess required by the detector.

Proof. Apply the certified energy theorem of [1] with $\kappa = 3/4$ and stage error $\rho/4$. Its positive-width and zero-slot conclusions give a positive fine mesh μ . The condition $\ell_j \leq \mu/200$ ensures $\lambda_j \leq \mu$ for every $d \geq 1/200$. This does not restrict the total slot length, since the number of distinct positive slots can be increased.

Fix the source's finite excluded prime set S , and let M_S be its product ideal. Use the fixed ideal $M_S \cap (72)$ in the energy source. For each target there are only finitely many source labels. The norm of the character modulus at every such label is at most $U^{\rho/4}$ once Z is large enough. The exceptional sixth-power-free rows are supported on a fixed finite set of primes and have bounded norm. They are therefore absent from the retained rows for large Z .

The source identities bound the sum of the squared plain witness products by retained energy with the physical prime coefficients. The marked initial state has width $1 + \rho/4$. Its capacity condition follows from $2m + 6\kappa L_A \leq 1$, and the positive-width energy bound gives exponent $1 + \rho/2 \leq 1 + \rho$. For the unmarked pair, use initial width $\max\{1, 2m\} + \rho/4$ and the zero-slot energy bound. This gives (6) and (7).

For the inverse witness, the fixed-source inverse estimates of [1] apply under the two strict capacity inequalities in (5). Distinct slot lengths give disjoint prime supports. Their fixed conductor deletions have modulus-one denominators on these supports for large Z . The inverse raw-moment bound and the marked inverse bound consequently apply to the perturbed slots.

The profile estimates bound all needed seminorms by a fixed power of $1 + H$. Taking a common exponent and summing the finitely many label constants gives J and C_η . The exponent is chosen before the target. Thus the estimates hold at detector height $H = Z^{2\tau}$,

and the usual late choice of $\tau > 0$ absorbs the polynomial height factor into the prescribed detector loss. $\square$

4 The high comparison

We first record the extension of the Euler regions used by the contour arguments. The local factors below are those of [1, Equations (5.13a)–(5.13c)].

Lemma 4. *The full compensated correction is holomorphic, with the bounds needed for principal-row contour moves, on the second Euler region of [1] with its condition $\operatorname{Re} s \geq 7/8$ replaced by $\operatorname{Re} s \geq B$. The principal local quotient there satisfies*

$$G_p/H_p = -1 + O((Np)^{-B}).$$

The small-row contour can use the first Euler region with $\epsilon_0 = 37/100$.

Proof. The exact local defect formula in [1, local Euler lemma] bounds $H_p - 1$ in the original second region by $O(Q^{-363/200})$ for $p \nmid u$ and $O(Q^{-33/40})$ for $p \mid u$, where $Q = Np$. Every local exponent involving $\operatorname{Re} s$ has coefficient of absolute value at most six. Moving its lower boundary left by $1/200000$ therefore gives the respective bounds

$$O(Q^{-363/200+3/100000}) \quad \text{and} \quad O(Q^{-33/40+3/100000}).$$

The first exponent is less than -1 . The unramified product converges normally, while the ramified product has the same $O_\varepsilon((Nu)^\varepsilon)$ bound. For $u = 1$ its tail still tends to zero faster than $P_0^{-4/5}$. Enlarging the fixed excluded set therefore gives the same uniform nonvanishing of the principal product.

The four errors for the principal selected factor have exponents $-\operatorname{Re} s$, $-6 \operatorname{Re} z$, $4 - 5 \operatorname{Re} s - 6 \operatorname{Re} z$, and $1 - \operatorname{Re} w - 6 \operatorname{Re} z$. On the extended region they are all at most $-B$. The formula for G_p has no division by P_p or H_p , so the full finite selected-tuple correction remains holomorphic at their zeros. The bounds above and finite slot count give the all-height majorant used in the principal contour proof of [1].

For small rows, $\beta_* > B$ implies $\beta_* + e + 1/2 > 1 + 37/100$. The first Euler region of [1] allows any positive ϵ_0 , so its unchanged local estimates apply to the line $(\beta_* + e, 1/2, 17/50)$. The central bin contours already lie in its region with $\epsilon_0 = 10e$. These are the only analytic domain uses of the original $7/8$ lower bound in the shared contour lemmas. Repeating their proofs with the extended region gives the same principal, bin, and outer-row contour estimates. $\square$

Lemma 5. *If $B < \beta_* \leq 7/8$, then for some $\sigma > 0$ independent of the target character,*

$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-\sigma},$$

where f_η is the principal Mellin signal from [1] with (2).

Proof. Lemma 3 supplies the fourth-moment estimate at $\kappa = 3/4$, because its zero-free input is $\beta_* \leq (1 + \kappa)/2 = 7/8$. Every detected bin has $\delta = 2a - 1 \leq 2\beta_* - 1 \leq 3/4$. Its plain prime capacity is thus the baseline capacity in the proof of [1, Part II, capacity comparison], with no positive $\Delta/4$ surcharge. The row exponent is bounded by $R_*(\delta, x)$ plus an arbitrarily small loss, with $0 \leq x \leq 1/2$ and $q = x\delta$. The same selected-prime argument applies since $\ell/h > 8/39 > 7/37$ and $5\ell - h > 1/48$. A fixed sufficiently fine even slot mesh meets all rounding conditions.

At row length $d = h$, the general high exponent relative to $C(B)$ is

$$E_B(h) = a - B + h \left(\frac{17}{50} - \frac{1}{6} \right) - al_y - \frac{\ell}{2} + q\ell + h \left(R + \frac{\delta}{2} - \frac{17}{50} \right). \quad (8)$$

The endpoint certificate of [1, Part II, compensated endpoint certificate], evaluated at its original geometry, bounds this expression with $B = 7/8$ and $R = R_*$ by $-49/440640$ for the larger range $\delta \leq 5/6$. Holding a, δ, q, R fixed while changing (1) gives

$$E_B(h) - E_{7/8}(13/16) = \frac{1}{200000} + t \left(-\frac{1}{2} + \delta + q + \frac{3R}{2} \right). \quad (9)$$

The source bounds $R_* \leq 17/12$. Since $\delta \leq 3/4$ and $q \leq \delta/2$, the right side of (9) is at most

$$\frac{1}{200000} + \frac{11t}{4} = \frac{7}{80000}.$$

Thus the central endpoint retains the explicit margin

$$-E_B(h) \geq \frac{49}{440640} - \frac{7}{80000} > 0. \quad (10)$$

The frequency slope is positive, so this endpoint bounds all moderate rows with $d \leq h$. Extending to $h + \zeta$ costs at most 2ζ , and ζ can be chosen below the margin in (10).

The floor and intermediate-row margins in [1] are respectively $7/1200$ and $49/14400$ at the original geometry. The floor shift costs $t(121/40 + 3e + \text{mesh})$, including the changed row threshold. The parameter budget absorbs this cost. The small-row margin is $63/800$. Its change is at most $51t/100$, since $h(17/50 - 1/6) - l_y/2$ has derivative $51/100$ with respect to t . For large rows, the absolute contour can still be taken sufficiently far right after choosing ζ . Lemma 4 supplies the principal contour, whose margins $l_y/20$ and $h/600$ remain positive. Its local approximation supplies a positive power saving in each fixed slot. The prime normalizer costs only $Z^{o(1)}$.

All detector, mesh, real-contour, and normalizer losses can be chosen below these fixed positive margins before the target character is fixed. The late height choice in [1] then gives one $\sigma > 0$ for every target, as claimed. $\square$

Proof of Theorem 1. Suppose $\beta_* > B$. Lemma 4 supplies a principal factor holomorphic and bounded away from zero on $\text{Re } s > B$. Lemma 2 and (4) give the direct bound required by the continuation criterion of [1, Section 2]. Its parameter ω can be any sufficiently small positive number below $\beta_* - B$. Lemma 5 gives the other required bound with a target-independent saving. The criterion contradicts $\beta_* > B$. Hence $\beta_* \leq B$. The transfer in [1] from primitive Hecke characters to imprimitive Hecke and Dirichlet characters proves the stated assertions. $\square$

Verification scope

The rational inequalities in (4) and (10) can be checked independently by the accompanying `scripts/verify_bounds.py`. The accompanying files in `lean/` verify the geometry, reflected-exponent calculation, high-side endpoint, normalized compensated low estimate, local Euler bounds, and principal and global Euler product extensions against the upstream OAI formalization. The finite source multiplier bounds and the principal source's w and z contour shifts, ordered double shift, uniform contour tails, and x contour transport also have Lean proofs at the new boundary. The selected high-row local factors,

weighted prime-slot sum, physical-row dyad, Mellin integral, and changed small- and large-row tail bounds, central cube truncation error, canonical ray-cube choice, and normalized floor-row collection are verified there as well. The principal slot-ratio and normalized residue estimates are proved on the new boundary. The physical principal remainder at the changed lengths is bounded on a fixed contour just right of $7/8$, leaving a power saving of $1/3000$ relative to β_* in the proposed interval. The perturbed source moment transport is proved in `OAIHighMomentTransport.lean`. It supplies the chosen fourth-moment input used by the central-bin estimate and detector assembly. `OAIHighUnconditional.lean` derives the prior $\beta_* \leq 7/8$ bound from the upstream unconditional module and concludes the stronger Hecke, Dirichlet, and zeta nonvanishing statements without a moment hypothesis. The final theorem’s axiom audit permits only the standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`. It rejects any other transitive axiom dependency, including incomplete proofs. The verification script requires the pinned upstream commit and a clean upstream working tree. The proof uses Lean 4.34.1 and upstream revision `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Compiling this note checks the TeX. The separate Lean build checks the proof. The stronger boundary in Theorem 1 is an extension of [1]’s stated theorem.

References

- [1] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\text{Re}(s) > 7/8$, 30 September 2026. Source consulted 7 October 2026. [Source TeX at the verified revision](#).